

IAS Frontiers Conference on

# **Geometry, Dynamics, and Learning**

GDL 2026

TALKS AND ABSTRACTS

Conference Booklet  
28-30 September 2026  
Nanyang Technological University, Singapore

# Equivariant Systems Theory

**Robert Mahony**

*Australian National University*

## Abstract

Equivariant Systems Theory is a relatively new field of nonlinear control that studies systems for which there is a transitive equivariant symmetry. The field differs from invariance as studied in geometric control where the symmetry is typically free and not transitive and the goal is to reduce the system to a trivial part (on the symmetry fibre) and study the interesting part (the dynamics on the base space). In Equivariant Systems Theory the symmetry is always transitive and often not free. The goal is to lift the system to symmetry group and use this structure to tackle problems such as state estimation (Equivariant Filter) and control (Equivariant Regulator).

In this talk, I present a taxonomy of equivariant systems and discuss the structure of these systems. I consider linear integrators, invariant systems on groups, linear systems, and group affine systems, and show how these well-known systems fit into the equivariant systems structure. I go on to define the class of fundamental systems that is one of the most exciting categories for future work. Finally, I define free equivariant and general equivariant systems to complete the taxonomy. If there is time, I will also discuss the equivariant structure of measurements and outputs.

## About the speaker

Robert Mahony is a Professor in the Research School of Engineering at the Australian National University. He received his BSc in 1989 in applied mathematics and geology, and his PhD in 1995 in systems engineering, both from the Australian National University.

He is a Fellow of the Australian Academy of Science, the Australian Academy of Technological Sciences and Engineering, and the IEEE. He served as President of the Australian Robotics Association from 2008 to 2011.

His research interests include nonlinear systems theory and its applications in robotics and computer vision. He is known for his work in aerial robotics, geometric observer design, matrix subspace optimisation, and image-based visual servo control.

# A Geometer's View on the Manifold Hypothesis

**Wilderich Tuschmann**

*Karlsruhe Institute of Technology*

## **Abstract**

At its core, the Manifold Hypothesis asserts that real-world, high-dimensional data is not uniformly or randomly distributed throughout its high-dimensional "ambient" space, but concentrated on or near a low-dimensional manifold (or a collection of manifolds) embedded in that high-dimensional ambient space. In my talk, I will discuss reasons and facts that speak for and against this fundamental supposition as well as alternatives from Riemannian and metric geometry.

# MMD-Regularized Unbalanced Optimal Transport

**Piyushi Manupriya**

*Indian Institute of Technology Hyderabad*

## **Abstract**

Unbalanced optimal transport extends classical optimal transport by relaxing the requirement that the two measures have equal total mass, enabling the comparison of unnormalized measures and providing greater robustness against noisy empirical measures. Most UOT formulations rely on  $f$ -divergence penalties, while the potential of integral probability metrics - particularly the maximum mean discrepancy (MMD) - remains relatively underexplored. This talk introduces a family of MMD-regularized unbalanced OT metrics and discusses their statistical and computational properties. It also shows how the corresponding UOT problem with sparsity constraints can be formulated as a weakly submodular maximization problem. The talk concludes with discussing ongoing explorations on dynamic formulations and some open problems.

# SPD Matrix Learning for Brain-Computer Interfaces and NeuroAI

Ce Ju

*The Chinese University of Hong Kong, Shenzhen*

## Abstract

Many representations of neural data are not simply Euclidean vectors, but possess intrinsic mathematical structures that should be considered during learning and analysis. For example, covariance matrices and functional connectivity representations derived from EEG and neuroimaging data can often be modeled as Symmetric Positive Definite (SPD) matrices, either naturally or through appropriate regularization. The space of SPD matrices forms a Riemannian manifold, whose geometric structure provides a principled foundation for representation learning and statistical analysis.

In this talk, I will review our exploration of SPD matrix learning over the past five years, with a focus on applications in Brain-Computer Interfaces (BCIs) and NeuroAI. By integrating Riemannian geometry, optimization, and deep learning, we have developed geometry-aware learning frameworks for neural representations to enable robust and effective neural decoding.

I will further discuss the extension of SPD-based approaches to neuroimaging analysis and functional-connectivity modeling, as well as our recent efforts in generative modeling of structured brain data. Together, these developments demonstrate how geometric modeling can bridge mathematical principles and artificial intelligence, providing new perspectives for advancing BCIs and NeuroAI.

# Invariant kernels on symmetric spaces

**Cyrus Mostajeran**

*Nanyang Technological University*

## **Abstract**

We develop harmonic-analytic foundations for the study of invariant positive-definite kernels on Riemannian symmetric spaces with applications including kernel construction on spaces of covariance matrices. New results provide practical criteria for positive-definiteness and universality in both compact and non-compact settings. Several examples and numerical experiments illustrate the theory and its applications to statistical learning problems.

# Kernel Learning of PDE Solution Operators

**Jianyu Hu**

*Nanyang Technological University*

## **Abstract**

A kernel-based approach for approximating the solution operator of linear nonhomogeneous partial differential equations (PDEs) is proposed. The method incorporates physical priors, encoded through the governing differential operator, into a kernel ridge regression framework to construct a closed-form operator learner. The resulting estimator induces a well-defined solution operator linking the input and solution spaces of the PDE, thereby shifting the perspective from solving individual PDEs to constructing an operator-based solver. In contrast to standard supervised learning methods, it does not rely on paired input-output training data and enables systematic extrapolation beyond observed regimes.

A full uniform operator-level convergence is conducted, providing convergence rates for the operator-based solver under suitable choices of regularization parameters. Numerical experiments demonstrate high accuracy and computational efficiency, while comparing favorably with Green operator learning in both approximation quality and computational cost.

# kNN: From a Topological Data Analysis (TDA) Perspective to a Topological Combinatorics Approach

**Minh Le Quang**

*Ho Chi Minh City Open University*

## **Abstract**

Topological Data Analysis (TDA) is a rapidly growing field of applied mathematics in which techniques from computational and applied topology are used to extract structural information about the “shape” of data. TDA has been applied in numerous contexts, ranging from data science and machine learning to biology and neuroscience, visualization and dimensionality reduction.

Topological Combinatorics is an area of mathematics that uses methods from topology to solve combinatorial problems. It studies how geometric and topological structures reveal information about discrete objects such as graphs, hypergraphs, simplicial complexes, and set systems.

In this work, the  $k$ -nearest neighbors (kNN) method is viewed as lying at the intersection of two complementary perspectives: topological data analysis (TDA) and combinatorics. From the standpoint of TDA, we investigate stability properties and establish convergence results. From the combinatorial perspective, we focus on the development of path homology

Path homology has emerged through a sequence of developments, beginning with combinatorial constructions and subsequently extending to a general categorical framework. In this work, we consider a class of dynamical paths constructed from the nearest-neighbor structure associated with each vertex. Within this setting, we define the corresponding path complex and its associated homology theory.

# Rank-one fluctuations in averaging-learning dynamics: agreement geometry without a ground truth

**Tushar Vaidya**

*Nanyang Technological University*

## Abstract

Averaging-learning dynamics describe a population that averages over a time-varying network while learning from a reference signal. We remove the usual exogenous ground truth: the reference is generated endogenously by the population itself. A Dobrushin-type contraction controls the decay of oscillations and yields asymptotic agreement; under a summability condition the backward products converge exponentially to rank-one limits. With summable perturbations the process converges to a random consensus state, and for i.i.d. perturbations we prove a central limit theorem whose Gaussian limit is supported on the agreement direction. Despite the multi-agent dynamics, the long-time fluctuations are asymptotically one-dimensional. A pairwise Dobrushin formulation exposes the underlying geometry as a dynamic partition into agreement classes, with the one-class regime being exactly where the rank-one picture holds.

# The analogy between ResNets and ODEs misses an expectation

**Lénaïc Chizat**

*National University of Singapore*

## **Abstract**

Residual networks are often described as Euler discretizations of ODEs. In this short talk, I will show that this analogy misses an important ingredient: for ResNets trained from standard random initialization, the vector field arising in the large-depth limit is given by an expectation. I will introduce a stochastic-approximation viewpoint on ResNets and explain how it clarifies the roles of depth and width in large-scale regimes.

# State-space and state-affine systems for learning from temporal data

**Florian Rossmannek**

*Nanyang Technological University*

## **Abstract**

State-space systems form a general prevalent modelling class for learning tasks in a temporal context, in particular for learning dynamical systems. We review the principles underlying the successful learning of dynamics with such models, and we discuss stability in time and synchronisation of dynamics, which we link to the notions of echo states and state forgetting. We review in more detail state-affine systems, a subclass that allures with simplicity yet turns out surprisingly expressive by trading complexity of the dynamics for dimensions. These state-affine systems have resurfaced recently in the context of quantum reservoir computing and sequential kernel methods.

# Timescale effects and dominant manifolds in reservoir computing networks

**Anastasia Bizyaeva**

*Cornell University*

## **Abstract**

Reservoir Computing (RC) is a gradient-free machine learning technique for data-driven modeling and prediction of temporal time series. In the presented work, we study RC through the lens of systems and control theory. Working in continuous time, we show that RC training can be interpreted as a data-driven feedback design problem, with feedback loops inducing low-dimensional dominant manifolds whose structure reflects the intrinsic dimensionality of the training data. We show how this systems-theoretic point of view naturally links RC to several classical techniques, including the Dynamic Mode Decomposition algorithm and Kazantzis-Kravaris-Luenberger observer design problems. The latter connection provides insight into reservoir learning via the observer internal model principle.

# Geometric adjoint sensitivity analysis with applications to neural ODEs and PDE constrained optimization

**Melvin Leok**

*University of California San Diego*

## Abstract

Adjoint systems are widely used to inform control, optimization, and design in systems described by ODEs or DAEs. At the level of PDEs, advection equations and mean field games also have the structure of adjoint systems, and all adjoint systems have a Hamiltonian structure.

We utilize symplectic and presymplectic geometry to investigate the properties of adjoint systems associated with ODEs and DAEs, respectively. We show that the adjoint-variational quadratic conservation laws, which are critical to adjoint sensitivity analysis, arise from (pre)symplecticity of such adjoint systems.

We develop structure-preserving numerical methods for such systems using Galerkin Hamiltonian variational integrators which admit discrete analogues of these quadratic conservation laws. This can be used to train neural networks arising from time-discretizations of neural ODEs, even ones based on implicit-in-time discretization.

Finally, we investigate the geometric structure of adjoint systems associated with evolutionary partial differential equations at the fully continuous, semi-discrete, and fully discrete levels and the relations between these levels. We show that the adjoint system associated with an evolutionary PDE has an infinite-dimensional Hamiltonian structure, which is useful for connecting the fully continuous, semi-discrete, and fully discrete levels.

## About the speaker

Melvin Leok is professor of mathematics and director of the Computational Science, Mathematics and Engineering program at the University of California, San Diego. His research interests are in computational geometric mechanics, computational geometric control theory, discrete differential geometry, and structure-preserving numerical schemes, and particularly how these subjects relate to systems with symmetry. He received his Ph.D. in 2004 from the California Institute of Technology in Control and Dynamical Systems under the direction of Jerrold Marsden. He was a PIMS Marsden Memorial Lecturer, Simons Fellow in Mathematics, three-time NAS Kavli Frontiers of Science Fellow, and has received the DoD Newton Award for Transformative Ideas, NSF Faculty Early Career Development (CAREER) Award, SciCADE New Talent Prize, SIAM Student Paper Prize, Leslie Fox Prize (second prize) in Numerical Analysis, A\*STAR International Fellowship, and Loke Cheng-Kim Foundation Scholarship. He has given plenary talks at Foundations of Computational Mathematics, NUMDIFF, and the IFAC Workshop on Lagrangian and Hamiltonian Methods for Nonlinear Control, and is the coauthor of a research monograph entitled, “Global Formulations of Lagrangian and Hamiltonian Dynamics on Manifolds.”

# Information-Geometric Port-Hamiltonian Reinforcement Learning

**Chandrajit Bajaj**

*The University of Texas at Austin*

## Abstract

I shall present an information-geometric port-Hamiltonian reinforcement-learning framework for stable geometric and scientific learning. Instead of learning unconstrained policy vector fields, the method learns stage-wise structured policies that select information actions, assimilate observations into memory, choose port-Hamiltonian templates, and advance cotangent phase states through energy-aware dynamics. Task divergences and optimal-transport losses define reward or residual potentials; Fisher-type measurements condition local safety, uncertainty, and pose/shape separation; and skew interconnection with positive semidefinite dissipation provides interpretable energy and power diagnostics. Adaptive B-spline deformation will serve as a geometric case study, where Sinkhorn-driven matching, fixed-period knot charts, measurement Fisher blocks, Souriau-Fisher pose statistics, active local support selection, and energy-audited integration combine into a robust policy for operator-constrained inverse shape search.

## About the speaker

Chandrajit Bajaj is the Computational Applied Mathematics Chair in Visualization, Professor of Computer Science, and Director of the Center for Computational Visualization at The University of Texas at Austin's Oden Institute. His research spans geometric modeling, imaging data science, visualization, computational biology, scientific computing, machine learning, and predictive medicine, with recent emphasis on stochastic dynamical processes, predictive decision-making, and variational machine learning. He received his B.Tech. from IIT Delhi and his M.S./Ph.D. from Cornell University. He is a Fellow of AAAS, ACM, SIAM, and IEEE, and a Distinguished Alumnus of IIT Delhi.

# Convolution, equivariance, and invariance in Dynamical Systems

**Ravi N. Banavar**

*Indian Institute of Technology Bombay*

## Abstract

The talk will present three fundamental and deeply interconnected constructs that lie at the heart of dynamical systems and signal processing: convolution, equivariance, and invariance. Invariance has long been rooted in mechanics and dynamical systems, where it enables profound simplifications of models and their associated computations. Equivariance, closely related, rather than enforcing strict constancy, captures how systems transform consistently under symmetry actions. It has played a central role across dynamical systems, observer design, and modern signal processing, where respecting the transformation structure of signals leads to architectures that are both efficient and robust. In recent years, equivariance has emerged as a guiding principle in the design of learning systems, particularly in ensuring that models generalize across structured variations in data. Convolution, historically entrenched in signal processing, provides the canonical operation for linear time- (or space-) invariant systems. It encodes locality, weight-sharing, and compositional structure, making it indispensable in both classical filtering theory and modern data-driven methods such as convolutional neural networks. More recently, convolution has undergone a conceptual expansion: from its classical formulation on Euclidean domains to more general settings on Lie groups. This shift brings symmetry to the forefront, allowing signal processing tools to operate on domains with intrinsic geometric and algebraic structure.

Together, these three constructs reveal a unifying theme: the central role of symmetry in shaping both physical systems and data-driven models. The talk will develop this connection, illustrating how invariance and equivariance naturally lead to convolutional structures, and how extending these ideas to group-theoretic settings opens new frontiers in signal processing and machine learning.

# **An invariant theoretic perspective on learning path dependences in finance and technology**

**Josef Teichmann**

*ETH Zurich*

## **Abstract**

We provide an invariant theoretic perspective on feature construction for learning problems and apply the insights to two prominent special cases: signature transforms and attention layers.

## **About the speaker**

Josef Teichmann, Professor at the Department of Mathematics of ETH Zurich since 2009, graduated from Vienna University in 1999 with a doctoral thesis on infinite dimensional Lie groups. His research interests include Mathematical Finance, Stochastic Analysis, and Machine Learning in Finance.

# Souriau's Dissipative Lie Groups Thermodynamics-Informed Machine Learning, Lee's Conformal Hamiltonian Dynamics & Kubo's Fluctuation-Dissipation extension

**Frédéric Barbaresco**

*THALES*

## Abstract

Souriau's Lie Groups Thermodynamics provides an advanced mathematical framework for unifying geometric statistical mechanics, information geometry, Lie-group machine learning, and physics-informed neural networks (PINNs). We apply Souriau's formalism to Thermodynamics-Informed Neural Networks (TINNs) endowed with a transverse foliation structure. The non-dissipative dynamics is defined on the symplectic leaves, interpreted as level sets of the entropy function, generated by coadjoint orbits. Dynamics strictly confined to these leaves is governed by the Poisson bracket and describes reversible, non-dissipative physical processes, for which entropy is an invariant Casimir function. The dissipative dynamics, by contrast, is defined transversally to the symplectic foliation, on the associated transverse Riemannian structure. Motion across distinct symplectic leaves represents irreversible processes and is driven by a metric bracket, as in metriplectic dynamics or the GENERIC framework. This formulation guarantees physically admissible, non-negative entropy production. We have identified links between this transverse symplectic foliation structure and Libermann foliation based on Pierre Molino and Pierre Dazord Works published in Sud-Rhodanien Seminar on "Geometry of Moment Map". In certain situations, notably those involving drag or frictional forces, the dissipative dynamics may be reduced to a conformally Hamiltonian dynamics by effectively freezing the entropy-production mechanism or treating the environment as an infinite thermal reservoir. In this limiting description, the metric bracket reduces to a simple linear friction vector field: the resulting metriplectic structure has a fixed, non-degenerate dissipative metric tensor and continuously transfers mechanical energy to an external, unobserved sink, without explicitly accounting for the associated temperature or entropy variations. A conformally Hamiltonian system with drag consequently violates the second degeneracy condition of the metriplectic formalism, since it represents an open system in which mechanical energy is irreversibly dissipated. If, however, the kinetic energy lost through drag is consistently transferred to the entropy or thermal sector of the metriplectic model, the resulting total system is isolated and conservative, even though its mechanical subsystem retains a conformally dissipative character. Hwa-Chung Lee initiated this model during 40's introducing flat symplectic cases alongside pairs of two-forms that scale conformally to each other, and laid the preliminary local framework for locally conformally symplectic. We finally consider an extension of Souriau's framework in connection with Callen's fluctuation-dissipation theory and Kubo's linear-response theory, a foundational pillar of modern statistical mechanics developed by the Japanese physicist Ryogo Kubo. Upon linearising the metriplectic equations in the vicinity of a thermal-equilibrium state, the components of the dissipative metric tensor become directly related to macroscopic transport coefficients, such as viscosity and thermal conductivity, which can be evaluated through the Green-Kubo relations. This establishes a natural geometric bridge between Souriau's Lie-group thermodynamics, metriplectic dynamics, irreversible thermodynamics, and microscopic linear-response theory.

## About the speaker

Frédéric Barbaresco is a French researcher and Principal Scientist at THALES who has extensively modernized and advanced Jean-Marie Souriau's Lie Groups Thermodynamics, a symplectic formulation of statistical mechanics, and extended its application to information geometry, Lie-group machine learning, and dissipative physics-informed neural networks. He has extended information geometry to Lie groups through the Koszul-Souriau-Fisher metric, introducing a dedicated Riemannian metric for the computation of Fréchet means, medians, and geodesic barycenter directly on coadjoint orbits. He has also proposed a geometric formulation of entropy as an invariant Casimir function defined on the symplectic foliation. His work on structure-preserving symplectic integrators translates sophisticated geometric theories into practical computational frameworks, exploiting (poly)symplectic geometry to ensure that fundamental physical structures, including thermodynamic principles, moment-map conservation, and underlying symmetries, are faithfully preserved throughout numerical integration. Through this program, Barbaresco establishes a bridge between abstract differential geometry and mathematical physics on the one hand, and demanding engineering applications on the other, including radar signal processing and flight-mechanics learning. General co-chair of SEE GSI (Geometric Science of Information) conference. Awarded Aymé Poirson Prize (2014) by the French Academy of Sciences to honor his work in applying science to industrial innovations.

# General Properties of the Thermodynamic Metrics for Generalized Souriau Partition Functions on CV manifolds

**Pietro Fré**

*University of Turin; INFN Turin*

## Abstract

Following previous results recently obtained by us, on Information Geometry versus Geometrical Thermodynamics and on the exact calculation of partition functions for extended Souriau Gibbs distributions on Calabi Vesentini manifolds, we study the differential geometry of the corresponding thermo-metrics. A general intriguing scheme is discovered and put into evidence. A small yet significant difference, distinguishes the even from the odd dimensional instance of the microscopic CV manifolds. Apart from that the complete thermo-space is flat when no constraint is introduced. Freezing the magnetic fields, which can be done according to complicated combinatorials, forces the thermo-system to evolve on curved submanifolds of the thermo-space that have a structure depending only on the length of the  $n - 1$  chain of frozen contiguous magnetic fields. The behavior of Riemann tensor components for such spaces is codified by a symmetric matrix with peculiar behavior along special symmetrically arranged submanifolds that, might be responsible for the generation of curvature walls and for the categorical partitioning of the thermo space. The embedding of this curved submanifold into  $\mathbb{R}^{2n-1}$  can be traced back to the vanishing of magnetic fields and, in this case, the flat metric on  $\mathbb{R}^{2n-1}$  is the  $\mathfrak{a}_{2n-1}$  simple Lie algebra Cartan matrix. In another version the flat embedding reveals the geometric interpretation of the  $n$ -manifold as a generalized translation hypersurface. The boundary at infinity has a hypercube structure whose face central points and vertices appear, numerically, to be the end-points of all geodesics depending only on their angular slope at the start. This general feature is reminiscent of the causal structure at infinity of Lorentzian space-times and of Penrose diagrams.

# A Universal Obstruction to the Samuelson Condition for Tangent Lagrangian 2-Webs

**Yasuhiro Kurokawa**

*Shibaura Institute of Technology*

## Abstract

The Samuelson area condition is a classical condition for Lagrangian 2-webs, with connections to integrability, flatness of canonical connections, thermodynamics, and economic web theory. In this talk, I will discuss a class of Lagrangian 2-webs in the symplectic plane generated by tangent lines to a regular plane curve.

The main result identifies a universal local obstruction to the Samuelson condition near the envelope of the tangent family. Geometrically, when two nearby tangent lines coalesce along the envelope, the Jacobian governing the web degenerates in a universal way; analytically, this gives rise to a logarithmic singularity in the Jacobian formulation of the Samuelson condition. It follows that near a point of nonzero curvature, the tangent Lagrangian 2-web associated with a regular plane curve cannot satisfy the Samuelson condition.

I will also briefly discuss possible connections with classical web geometry, bi-Lagrangian geometry, and related questions of separability, integrability, and flatness of canonical connections.

# A Geometric Perspective on Asymmetric Relaxation Dynamics

**Alessandro Bravetti**

*University of Camerino*

## **Abstract**

Asymmetric relaxation toward equilibrium is a recently-identified phenomenon that has attracted increasing attention in the study of thermodynamic systems. In this talk, we introduce a geometric approach based on gradient flows on appropriate manifolds to describe relaxation dynamics. This framework allows us to formulate a general criterion for determining the emergence of asymmetry in the approach to equilibrium. We conclude by outlining future perspectives and potential extensions.

# Beyond PINNs: From Natural-Gradient Geometry to Hybrid Galerkin Methods

**Nilo Schwencke**

*Inria; ENS de Lyon; CNRS*

## Abstract

Physics-informed neural networks and finite-element methods are commonly presented as fundamentally different paradigms: the former as nonlinear residual minimization, and the latter as variational projection onto a fixed approximation space. This talk develops a common geometric and kernel-theoretic framework for both.

A parametrized neural solver defines a finite-dimensional manifold in function space, and natural-gradient training amounts to projecting the functional residual onto its evolving tangent space. By contrast, a finite-dimensional Galerkin space can be viewed as a flat model manifold and, when equipped with the energy metric, as a reproducing-kernel Hilbert space whose kernel represents a projected Green operator.

This viewpoint leads to a reinterpretation of natural-gradient and Gauss-Newton approaches to PINNs as Petrov-Galerkin methods for the linearized PDE residual, with test functions implicitly induced by the geometry of the neural tangent space. Once the test space is treated as a design variable, strong collocation can be replaced by weak and operator-adapted tests. We exploit this freedom to construct a hybrid finite element-neural network in which the finite element component enforces the equation on a stable approximation space, while the neural network is trained only in its energy-orthogonal complement. Numerical experiments on multiscale and nonsmooth elliptic problems illustrate how this decomposition combines the stability of Galerkin methods with the adaptivity of neural tangent spaces.

The talk concludes with a discussion of residual-reconstruction criteria for adapting the effective tangent rank and regularizing natural-gradient training.

# Lie-Poisson Neural Network with control and dissipation for multi-rotor flight mechanics

**Meng Wu**

*Nanyang Technological University*

## **Abstract**

We propose to extend Lie-Poisson Neural Networks (LPNets) to model the controlled and dissipative dynamics of quadcopters in the reduced Lie - Poisson formulation. The approach aims to retain the intrinsic Lie-Poisson structure of the conservative dynamics while incorporating non-conservative effects in a physically consistent way. Using ground-truth data generated in PyBullet, we successfully learned the underlying quadcopter dynamics with high fidelity.

# Structured preserving Lagrangian neural methods for Vlasov equations with turbulence

**Emmanuel Franck**

*Inria; University of Strasbourg; CNRS*

## **Abstract**

The numerical solution of the Vlasov equation is challenging because it can generate turbulence in high-dimensional phase space. Classical methods require a vast number of degrees of freedom, leading to substantial computational and memory costs. Here, we propose to study a hierarchy of neural approaches to the Vlasov equation, ultimately focusing on Lagrangian methods that respect its geometric structure and capture turbulent dynamics.

# Learning, approximation and control

**Qianxiao Li**

*National University of Singapore*

## Abstract

In this talk, we discuss some interesting problems and recent results on the interface of deep learning, approximation theory and control theory. Through a dynamical system viewpoint of deep residual architectures, the study of model complexity in deep learning can be formulated as approximation or interpolation problems that can be studied using control theory, but with a mean-field twist. In a similar vein, training deep architectures can be formulated as optimal control problems in the mean-field sense. We provide some basic mathematical results on these new control problems that so arise, and discuss some applications in improving efficiency, robustness and adaptability of deep learning models.

## About the speaker

Qianxiao Li is an associate professor in the Department of Mathematics, and a principal investigator in the Institute for Functional Intelligent Materials, National University of Singapore. He graduated with a BA in mathematics from the University of Cambridge and a PhD in applied mathematics from Princeton University. His research interests include the interplay of machine learning and dynamical systems, control theory, stochastic optimisation algorithms and data-driven methods for science and engineering.

# Geometric Integration and Its Applications to Optimization and Machine Learning

**David Martín de Diego**

*Instituto de Ciencias Matemáticas (ICMAT-CSIC)*

## Abstract

Geometric numerical integration provides structure-preserving algorithms that accurately reproduce the qualitative features of dynamical systems over long time intervals. In this talk, we will illustrate how geometric methods, and in particular symplectic integrators and discrete variational calculus, offer powerful computational tools for a wide range of modern applications.

We will present three representative examples. First, we will discuss fuel-optimal navigation problems, where a parallel iterative variational method leads to efficient trajectory optimization. Next, we will show how geometric ideas can be used to accelerate gradient-based optimization algorithms for minimizing loss functions arising in machine learning. Finally, we will consider Hamiltonian learning from data, highlighting how symplectic integration naturally preserves the underlying physical structure and improves the quality of the learned models.

These examples demonstrate how ideas originating in geometric mechanics and variational principles can be successfully transferred to applied problems in optimization, control, and scientific machine learning.

## About the speaker

David Martín de Diego is a Research Scientist at the Instituto de Ciencias Matemáticas (ICMAT), Spain. His research focuses on the development and application of geometric mechanics to a wide range of areas, including geometric numerical integration, optimization, machine learning, nonholonomic mechanics, Hamilton-Jacobi theory, optimal control, Lagrangian reduction, robotics, multi-agent systems, and the interconnection of mechanical systems.

His work emphasizes the interplay between differential geometry - particularly symplectic, Poisson, and Dirac geometry - and the qualitative analysis of continuous and discrete dynamical systems, with applications to mechanical systems, systems with symmetry, and optimal control.

He serves on the Editorial Boards of the Journal of Nonlinear Science, Geometric Mechanics, the Mediterranean Journal of Mathematics, and La Gaceta de la Real Sociedad Matemática Española (RSME), where he served as Editor-in-Chief from 2003 to 2008.

# Beyond Symmetry: Efficient, Generalizable Learning for Weakly Invariant Systems

**Jake Welde**

*Cornell University*

## Abstract

Symmetry has long played a role in enabling high-dimensional robotic systems to operate autonomously in complex environments and under previously unseen conditions, despite stringent computational constraints. Nonetheless, the classical notion of symmetry is too rigid to be applied to many real-world robotic systems, where symmetry may be “broken” by external forces or confined to only a portion of the overall system. To overcome these limitations, we propose a relaxed notion of symmetry - termed “weak invariance” - that balances structure with generality. We show that weak invariance is in fact a necessary and sufficient condition for the reduction of control problems pertaining to relative motion, and we leverage these reductions for sample-efficient learning of tracking controllers for robotic systems.

# Synchronous Models and Fundamental Systems in Observer Design

**Pieter van Goor**

*University of Sydney*

## Abstract

Equivariance- and invariance-based observers and filters have seen increasing impact in the state estimation literature over the past decades, particularly for applications in robotics, navigation, and aerospace. A common feature of those observer designs that exhibit (almost-)global stability is that the uncorrected error dynamics remain stationary, allowing Lyapunov analysis to be undertaken without the need to compensate or dominate exogenous terms introduced by the input signal. This property is formally defined as the observer containing a synchronous model, and is an extension of the internal model concept used in observer design for dynamical systems. A system is said to contain a synchronous model of another if there is a suitable error function between the two systems that remains stationary for all of the trajectories of the two systems. For any given system, we provide necessary and sufficient conditions that there exists another system that contains a synchronous model of the first, and a method to construct a (there may be many) model should one exist. We characterise the class of all systems for which synchronous models can be constructed by showing that they consist of fundamental vector fields induced by a Lie group action, a class of systems we term fundamental systems. For fundamental systems we propose a simple synchronous observer design methodology, for which we show how correction terms can be discretised and combined easily, facilitating global characterisation of convergence and performance. Finally, we provide examples to demonstrate the key concepts of synchrony, symmetry construction, and observer design for a fundamental system.

# Generalized Homogeneity and Its Applications

**Andrey Polyakov**

*Inria; University of Lille; CNRS*

## Abstract

The generalised homogeneity is a scaling (dilation) symmetry, which can be discovered in some models of physical systems. A non-standard dilation allows the original state space of the system to be equipped with a new rule for multiplication of a vector by a scalar, with a new rule for addition of vectors, with a new norm (in the case of Banach space) and with a new inner product (in the case of Hilbert space). A linear controller in this new space corresponds to a generalised homogeneous nonlinear feedback law in the original space. Homogeneous controllers may demonstrate faster convergence, better robustness and improved safety comparing with linear algorithms. The declared theoretical advantages of homogeneous controllers have been confirmed by series experiments, which are going to be reported in this talk. Possible applications of the generalised homogeneity in machine learning are going to be briefly discussed.

# Statistical Potentials and Inverse Problems in Information Geometry: A Bi-Tensorial Approach

**Marco Pacelli**

*Scuola Superiore Meridionale; University of Naples Federico II*

## Abstract

Contrast functions and divergences play a central role in information geometry and in classical and quantum estimation theory. Geometrically, their diagonal jets generate statistical manifolds, namely metric-affine structures with conjugate torsion-free affine connections, and every statistical manifold arises from such a potential. Natural structures in quantum information geometry also lead to dual affine connections that need not be torsion-free, motivating the study of more general potentials depending also on tangent directions, such as pre-contrast and super-contrast functions. In this talk, the generation and inverse problems for these metric-affine structures are addressed by introducing suitable bi-forms within the bi-tensorial formalism. These bi-forms are equivalent to super-contrast functions and provide geometric potentials for structures with torsion. Via the natural commutative bicomplex on bi-forms, contrast and pre-contrast functions are recovered as bi-exact and partially exact potentials, respectively. The same framework yields a constructive solution to the inverse problem via vector transporters (or parallelisms) and recovers the Ay-Amari and Henmi-Kobayashi potentials via Poincaré-type constructions.

# New faces of Bregman divergences

**Frank Nielsen**

*Sony Computer Science Laboratories*

## Abstract

In this talk, we first show how rewriting Bregman divergences as power distances between weighted points using the mixed primal-gradient parameterization allows one to recast the Bregman minimum enclosing ball (MEB) problem as an equivalent power MEB problem (2607.24197). We then show how to solve power MEBs exactly using a LP-type recursive algorithm or approximately with arbitrary fine precision with the Frank-Wolfe optimization framework. Second, we present a generalization of Bregman divergences in Hilbert spaces based on the Riesz representation theorem yielding kernelized Bregman divergences that generalize the maximum mean discrepancy (2604.24047). Finally, we consider a generalization of polarity that introduces gauge freedom and a rewriting of Bregman divergences as inner products. This representation can be evaluated efficiently on GPUs and provides the foundation for neural network architectures that approximate the Legendre-Fenchel transform and neural information geometry (2603.04812).

# Bayesian Learning at Scale: Relaxations, Duality and Algorithms

Thomas Möllenhoff

*RIKEN*

## Abstract

Bayesian learning provides a framework for uncertainty-aware training of deep neural networks but traditionally has been difficult to scale to large models. The talk discusses several connections between optimization, duality, Bayes and recent algorithms for large-scale deep learning. In particular, we show how sharpness-aware minimization can be viewed as a dual relaxation of the Bayesian learning objective. This view motivates our Improved Variational Online Newton (IVON) and Eigenspace Variational Online Newton (EVON) algorithms that entail only small modifications to contemporary deep learning optimizers such as Adam or SOAP but optimize a variational Bayesian objective. In the last part of the talk, I show how primal-dual algorithms such as the Alternating Direction Method of Multipliers (ADMM) can be extended to the variational Bayesian setting, leading to new and improved methods in federated deep learning.

# Maxitive Donsker-Varadhan Formulation for Possibilistic Variational Inference

Jérémy Houssineau

*Nanyang Technological University; University of Warwick*

## Abstract

Variational inference (VI) is a cornerstone of modern Bayesian learning, enabling approximate inference in complex models. However, its formulation depends on expectations and divergences defined through high-dimensional integrals, often rendering analytical treatment impossible and necessitating heavy reliance on approximations. Possibility theory, an imprecise probability framework, allows us to directly model epistemic uncertainty instead of relying on a subjective interpretation of probabilities. While this framework provides robustness and interpretability under sparse or imprecise information, adapting VI to the possibilistic setting requires rethinking core concepts such as divergences, which presuppose additivity. In this work, we develop a principled formulation for performing possibilistic VI by establishing a maxitive analogue of the classical Donsker-Varadhan formulation. The resulting framework enables us to derive a learning rule for possibilistic VI with exponential-family candidates and practical update rules for neural-network training, giving rise to a family of optimizers termed CBOpt. Finally, we demonstrate that CBOpt achieves competitive performance on both in-domain and out-of-domain image classification tasks.

# Constructive Disintegrations and Conditional Modes

**Nathaël da Costa**

*University of Tübingen*

## Abstract

Conditioning, the central operation in Bayesian statistics, is formalised by the notion of disintegration of measures. However, due to the implicit nature of their definition, constructing disintegrations is often difficult. A folklore result in machine learning conflates the construction of a disintegration with the restriction of probability density functions onto the subset of events that are consistent with a given observation. We provide a set of mathematical tools which can be used to construct disintegrations and apply these to find densities of disintegrations on differentiable manifolds. Using these, we showcase a disturbingly simple example in which the restricted density and the disintegration density drastically disagree, due to a geometric term. Motivated by applications in approximate Bayesian inference, we further study the modes of disintegrations. We show that MAP estimation is ill-defined in this setting, as it yields modes of the restriction instead of the disintegration.

# Information flow, its geometry and information inequalities in stochastic flows: From uncertainty quantification in data-driven state estimation of non-autonomous dynamics to machine learning.

**Michal Branicki**

*Great Bay University*

## **Abstract**

I will present an overview of several themes unified by a consideration of a flow of information in data-driven, time-dependent state estimation of partially observed systems based on streams of noisy and incomplete data. These issues can be thought of as learning and training under uncertainty, and such an approach blends information theory and stochastic analysis within a probabilistic, differential-geometric framework complemented by Bayesian learning theory. The goal is to systematically understand and mitigate the key obstacles to robust and accurate prediction of finite- and infinite-dimensional dynamical systems in the presence of a practically inevitable modelling errors. I will outline the importance of certain information inequalities and their use in path-based predictions, data assimilation, and in the linear response theory. It turns out that similar techniques can be utilised to analyse the approximating capacity and robustness of estimates derived from neural networks. The talk will be largely accessible to a general mathematical audience with more technical details, time permitting, postponed to the final part.

# From stochastic dynamics to agentic learning equations, and toward AI scaling

**Lancelot Da Costa**

*Max Planck Institute for Intelligent Systems*

## **Abstract**

In this talk, we review recent advances in stochastic processes, focusing on Bayesian mechanics - a statistical-physics framework that links dynamics across informational boundaries to Bayesian inference and decision-making. This framework provides equations of motion for describing physical agents and, in turn, for constructing artificial agents that perform active Bayesian inference. We then examine the principal challenges and recent efforts in scaling this approach within AI. These challenges are fundamentally statistical: defining universal, searchable spaces of hierarchical Bayesian world models and performing inference across heterogeneous model spaces. Relevant to this is the information-geometric problem of measuring distances between probability distributions defined on different spaces.

# Policy Gradient Descent for Stochastic Control

**Zhenjie Ren**

*Université Évre Paris-Saclay*

## Abstract

Reinforcement learning has achieved remarkable success in landmark applications such as AlphaGo and AlphaFold, and is increasingly expected to play a central role in emerging domains such as autonomous driving. A substantial body of work has established theoretical foundations for classical reinforcement learning algorithms, including temporal-difference learning, policy iteration, and Q-learning. By contrast, policy gradient descent, despite its widespread use, remains less well understood from a theoretical perspective, largely because of its non-convex structure and the complexity of the underlying functional space. Existing convergence analyses typically require uniform regularity assumptions on the policy, viewed as a feedback control function, along the gradient flow; moreover, the resulting convergence rates depend on these regularity bounds. In this work, we significantly strengthen the existing convergence theory. Our key insight is to relate policy gradient descent to a mirror flow on the space of probability measures over controlled trajectories, where the Bregman divergence is induced by the convex control cost. We rigorously prove a JKO-type convergence result showing that the discrete-time mirror flow converges to its continuous-time counterpart, and we identify the limiting dynamics as a preconditioned policy gradient descent flow on the space of control processes. Leveraging this mirror-flow perspective, we establish an exponential convergence rate that is independent of policy regularity. The convergence holds both in Bregman divergence and in the value of the associated control problem.

# New quantum algorithms based on Riemannian optimization

**Marek Gluza**

*Nanyang Quantum Solutions; Nanyang Technological University*

## Abstract

I will show how basic aspects of the Riemannian geometry of the unitary group such as its curvature, tangent space and dense covering can guide the design of quantum algorithms. For this, we will derive an explicit formula for tangential gradients of linear cost functions which will show that corresponding state updates are in form of exponentials of commutators. Such unitary operations can be systematically implemented on quantum computers using the framework of double-bracket quantum algorithms which recently resulted in generalizations of the paradigmatic Grover's algorithm and new unitary synthesis formulas for implementing quantum signal processing. I will summarize the results of an ongoing collaboration developing this approach and I hope that at the end of the talk you will feel that knowing linear algebra is enough to take up designing quantum algorithms yourself.

# Deep learning over Riemannian manifolds

**Ziheng Chen**

*University of Trento; Max Planck Institute for Intelligent Systems*

## **Abstract**

Manifold-valued data arise in many machine learning applications, motivating the development of Riemannian deep learning. However, designing neural network layers that respect the diverse geometries of these manifolds remains challenging. In this talk, I will present two complementary approaches. The algebraic approach uses Lie group and gyrogroup structures to generalize batch normalization and attention. The geometric approach reinterprets classical classification layer through geometric concepts, leading to classifiers for Riemannian networks. Together, these examples show how algebraic and geometric structures can guide the design of Riemannian neural networks.

# Shape Regression and Learning: Distributional Robustness and Gradient Flows

**Yong Sheng Soh**

*National University of Singapore*

## **Abstract**

A broad range of learning problems can be abstractly viewed as performing regression of a suitably parameterized function to data. When these functions are viewed via its level sets, these learning tasks can be equivalently viewed as instances of shape regression tasks. In this talk, we will examine learning tasks via the perspective of shape regression tasks. This allows us to use geometric ideas to understand phenomena that are of interest from a practitioner's perspective. We characterize the optimal level sets given the underlying data distribution in settings where the level sets are required to be convex bodies as well as star bodies. We discuss how incorporating distributional robustness (in our learned model) affects the optimal level sets. Finally, we discuss gradient flows over the space of shapes we learn as a means of understanding the behavior of learning algorithms.